

Physical Modeling and Information-Theoretic Analysis of Holographic MIMO Channels

Workshop on Communication Technology at TU Berlin

by Dr. Andrea Pizzo
12 May 2025

Outline

- Holographic MIMO
- Physical Modeling of Multipath Fading
- Modeling Coupling as a Spatial Filtering Operation
- Reconfigurable Antenna Arrays
- Conclusion

Holographic MIMO

Electrically Large Arrays

- MIMO evolves by **expanding array apertures** compared to the wavelength λ
- Larger apertures enable finer sensing and additional spatial dimensions for communications
- Antennas are typically **spaced at $\lambda/2$** to ensure statistical independence and minimize mutual coupling
- With fixed spacing, large arrays require **more antennas** or **shorter wavelengths**
- However, physically large arrays are hard to implement and deploy
- And short wavelengths suffer from blockage, reduced range, lower scattering richness

Definition

- Further progress in large arrays are attained by shrinking antenna spacing below $\lambda/2$
- New multiantenna paradigm defined holographic MIMO (HMIMO)
- “*Holography*” has Greek roots and means “*describe everything*”, reflecting the aim to manipulate the channel entirely¹
- Signal theory approach to model electromagnetic phenomena such as multipath fading and mutual coupling

Physical Modeling of Multipath Fading

Spectral Representation of Random Processes

- Complex-Gaussian stationary random process $h(t)$ of baseband bandwidth B [1/s]
- Correlation function $r(t) = E\{h(\tau) h^*(\tau+t)\}$ is Hermitian and positive semidefinite
- Spectral theorem ensures existence of a real and non-negative power spectrum $S(f)$

$$r(t) = \int_{-B}^B S(f) e^{j2\pi f t} df, \quad -\infty < t < \infty$$

- **Spectral representation** via a complex-Gaussian, statistically independent random spectrum with power density $S(f)$ ¹

$$h(t) = \int_{-B}^B W(f) \sqrt{S(f)} e^{i2\pi f t} df, \quad -\infty < t < \infty$$
$$W(f) \sim \mathcal{N}_c(0, 1) \text{ IID}, \quad |f| \leq B$$

Asymptotic Karhunen-Loeve Expansion

- Observe $r(t)$ over a duration T and repeat the function to make it periodic
- The power spectrum becomes a train of impulses separated by $1/T$ (Fourier series)

$$r(t) \approx \sum_{i=1}^{2BT} S(f_i) e^{j2\pi f_i t}, \quad f_i = \frac{1}{T} i$$

- When $BT \gg 1$, $r(t)$ is approximated by enough samples
- Asymptotic Karhunen-Loeve expansion¹

$$h(t) \approx \sum_{i=1}^{2BT} w_i \sqrt{\lambda_i} \phi_i(t), \quad -\frac{T}{2} < t < \frac{T}{2}$$

$$w_i \sim \mathcal{N}_{\mathcal{C}}(0, 1) \text{ IID}, \quad \lambda_i = T S(f_i), \quad \phi_i(t) = \frac{e^{j2\pi f_i t}}{\sqrt{T}}$$

From Signals to Electromagnetic Fields

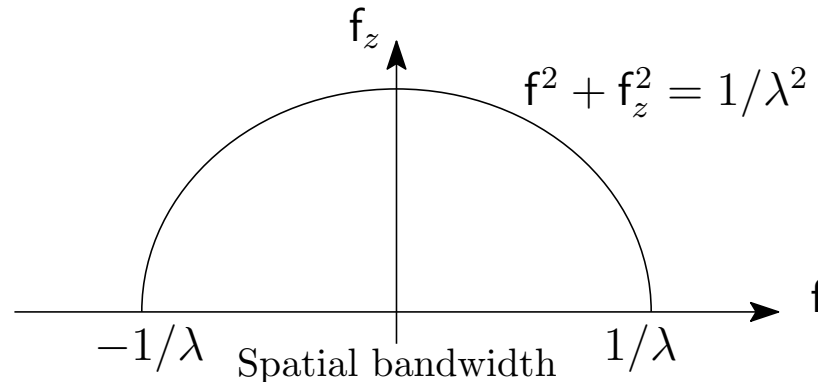
- Complex-Gaussian stationary electromagnetic random field $h(x,z)$ (spatial bandwidth?)
- Correlation $r(x,z)$ satisfies the Helmholtz equation¹ (linear and space-invariant operator)

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} + \frac{(2\pi)^2}{\lambda^2} \right) r(x, z) = 0, \quad (x, z) \in \mathbb{R}^2$$

- Eigenfunctions are complex exponentials (plane waves) with 2D spatial frequency (f, f_z)

$$\phi(x, z) = e^{j2\pi(fx + f_z z)}, \quad f^2 + f_z^2 = 1/\lambda^2$$

- Parametrization over f gives the spatial bandwidth $1/\lambda$



Spectral Representation of Electromagnetic Random Fields

- **Spectral representation** via a complex-Gaussian, statistically independent random spectrum with power density determined by **environmental scattering**¹
- Due to circularly symmetry of complex-Gaussian variables, **z translations are immaterial**

$$h(x, \mathbf{z}) = \int_{-1/\lambda}^{1/\lambda} W(\mathbf{f}) \sqrt{S(\mathbf{f})} e^{i2\pi\mathbf{f}x} e^{i2\pi\mathbf{f}\mathbf{z}(\mathbf{f})\mathbf{z}} d\mathbf{f} \sim h(x), \quad -\infty < x < \infty$$

$$W(\mathbf{f}) \sim \mathcal{N}_c(0, 1) \text{ IID}, \quad S(\mathbf{f}) = \frac{E\{|H(\mathbf{f})|^2\}}{\sqrt{1/\lambda^2 - |\mathbf{f}|^2}}, \quad |\mathbf{f}| \leq 1/\lambda$$

Asymptotic Karhunen-Loeve Expansion of Electromagnetic Random Fields

- Observe $r(x)$ over a segment of size L and repeat the function to make it periodic
- The power spectrum becomes a train of impulses separated by $1/L$

$$r(x) \approx \sum_{i=1}^{2L/\lambda} \left(\int_{f_i}^{f_{i+1}} S(\xi) d\xi \right) e^{j2\pi f_i x}, \quad f_i = \frac{1}{L} i$$

- When $L/\lambda \gg 1$, $r(x)$ is approximated by enough samples
- Asymptotic Karhunen-Loeve expansion¹

$$h(x) \approx \sum_{i=1}^{2L/\lambda} w_i \sqrt{\lambda_i} \phi_i(x), \quad -\frac{L}{2} < x < \frac{L}{2}$$

$$w_i \sim \mathcal{N}_{\mathcal{C}}(0, 1) \text{ IID}, \quad \lambda_i = L \int_{f_i}^{f_{i+1}} S(f) df, \quad \phi_i(x) = \frac{e^{j2\pi f_i x}}{\sqrt{L}}$$

Physics-Informed Fourier Model

- **Sampling** the asymptotic expansion at antenna locations yields the Fourier model¹
- Isometric decomposition in terms of **discrete Fourier transform matrices**
- Extends the **virtual channel model** to near-field propagation²

$$\mathbf{H} = \mathbf{U} \tilde{\mathbf{H}} \mathbf{V}^H$$
$$[\mathbf{U}]_{n,i} \approx \frac{e^{j \frac{2\pi}{N_r} i n}}{\sqrt{N_r}}, \quad [\tilde{\mathbf{H}}]_{i,\ell} \sim \mathcal{N}_c \left(0, \frac{N_r N_t}{L_r L_t} \lambda_{i\ell} \right) \text{ IID}, \quad [\mathbf{V}^H]_{\ell,m} \approx \frac{e^{-j \frac{2\pi}{N_t} \ell m}}{\sqrt{N_t}}$$


$N_r \times 2L_r/\lambda$ $2L_r/\lambda \times 2L_t/\lambda$ $2L_t/\lambda \times N_t$

[1] A. Pizzo, L. Sanguinetti, T. Marzetta, "Fourier Plane-Wave Series Expansion for Holographic MIMO Communications," IEEE Trans. Wireless Commun., 2022.

[2] A. Sayeed, "Deconstructing multiantenna fading channels," in IEEE Trans. Sig. Process., 2002.

Physics-Informed Fourier Model

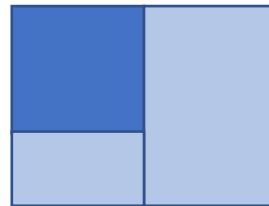
- **Sampling** the asymptotic expansion at antenna locations yields the Fourier HMIMO model¹
- Isometric decomposition in terms of **discrete Fourier transform matrices**
- Extends the **virtual channel model** to near-field propagation
- **Low-rank approximation** of the unitary-independent-unitary (UIU) model¹

$$\mathbf{H} = \mathbf{U} \tilde{\mathbf{H}} \mathbf{V}^{\text{H}}$$

$$\mathbf{U} = \text{Eig}(\mathbf{H} \mathbf{H}^{\text{H}}), \quad [\tilde{\mathbf{H}}]_{i,\ell} \sim \mathcal{N}_{\mathcal{C}}(0, \lambda_{i\ell}(\mathbf{R}_{\mathbf{H}})) \text{ IID}, \quad \mathbf{V} = \text{Eig}(\mathbf{H}^{\text{H}} \mathbf{H})$$



$$N_{\text{r}} \times N_{\text{r}}$$

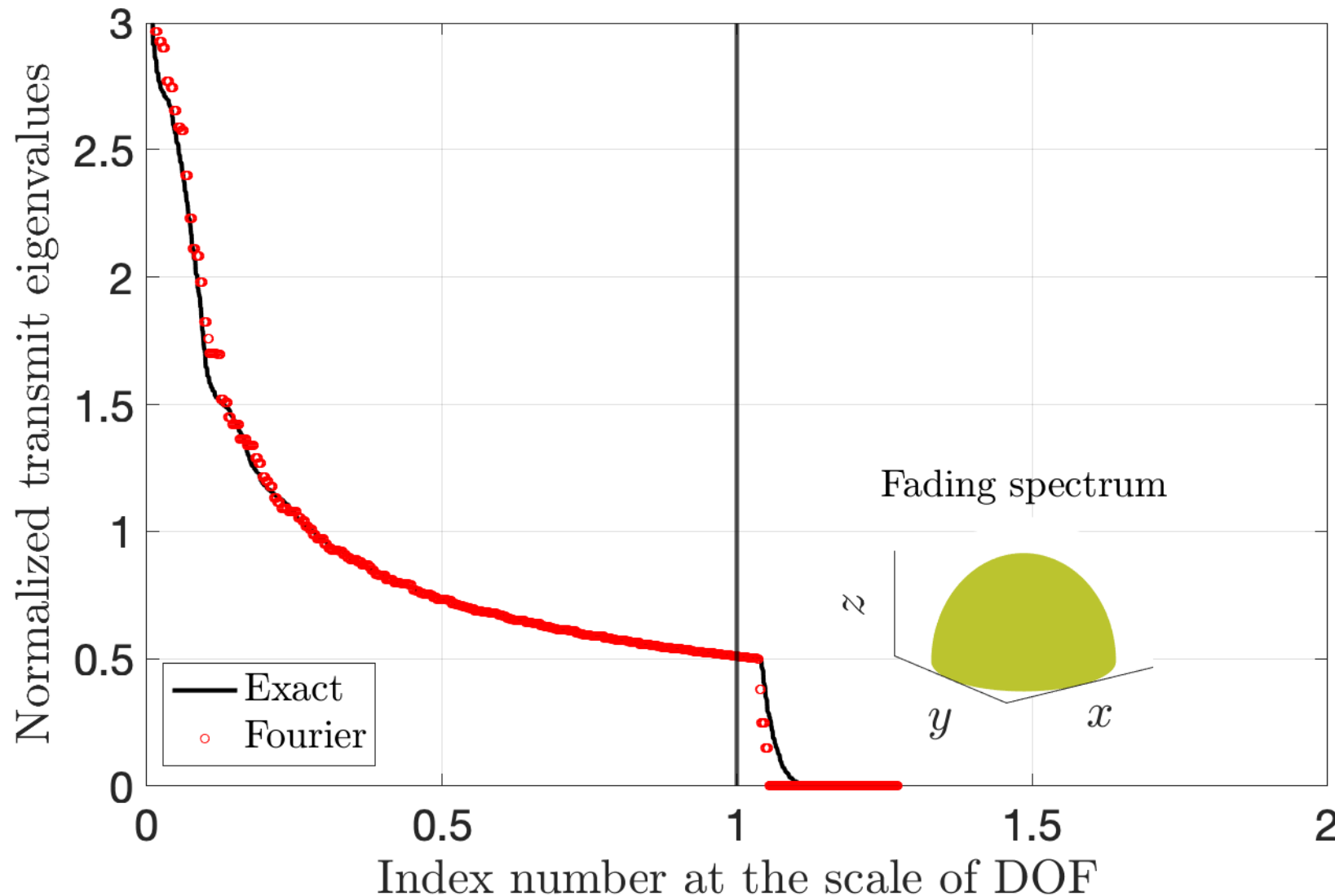


$$N_{\text{r}} \times N_{\text{t}}$$



$$N_{\text{t}} \times N_{\text{t}}$$

Fourier Model Accuracy with Finite Apertures



$$L_t/\lambda = 20$$

$$d_t/\lambda = 1/4$$

Separable scattering

Clarke – Jakes spectrum

$$\text{DOF} = \left\lceil \frac{2L_t}{\lambda} \right\rceil$$

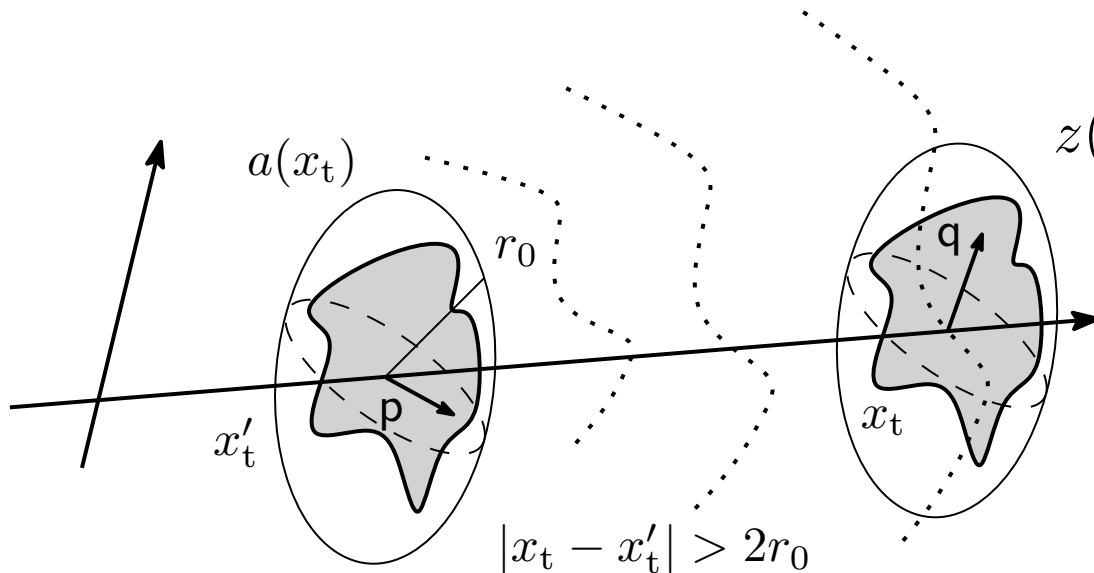
Modeling Coupling as a Spatial Filtering Operation

Impedance Function

- Similar to fading, **mutual coupling** can also be **modeled as a spatial filter**¹
- The **filter response** is given by the impedance function $z(x_t, x'_t)$

$$v_t(x_t) = \int_{-\infty}^{\infty} z(x_t, x'_t) j_t(x'_t) dx'_t$$

- For **physical antennas**, the impedance results in a convolution (spectrum multiplication)



$$z(x_t - x'_t) = \int_{-\infty}^{\infty} dp a(p - x'_t) \int_{-\infty}^{\infty} dq G(p - q) a(q - x_t)$$

$$Z(f) = \frac{|A(f)|^2}{\sqrt{1/\lambda^2 - |f|^2}} \quad |f| \leq 1/\lambda$$

Multipath Fading and Mutual Coupling

- Fading and coupling can be regarded as **spatial filtering operations**¹
- However, fading maps directly onto the channel while coupling applies at the antenna level
- Putting both phenomena on equal footing requires a **deconvolution (inverse spectrum)**
- **Antenna power pattern** acts as a proxy for coupling

$$h_Z(x) = \int_{-1/\lambda}^{1/\lambda} W(f) \sqrt{S_Z(f)} e^{i2\pi f x} df, \quad -\infty < x < \infty$$

$$S_Z(f) = \frac{E\{|H(f)|^2\}}{|A(f)|^2}, \quad |f| \leq 1/\lambda$$

Physics-Informed Fourier Model with Coupled Antennas

- Approximating the integral representation with the series expansion and sampling at antenna locations yields the Fourier model with **coupled antennas**¹
- Coupling enters the Fourier model only through the **variances of the independent matrix**

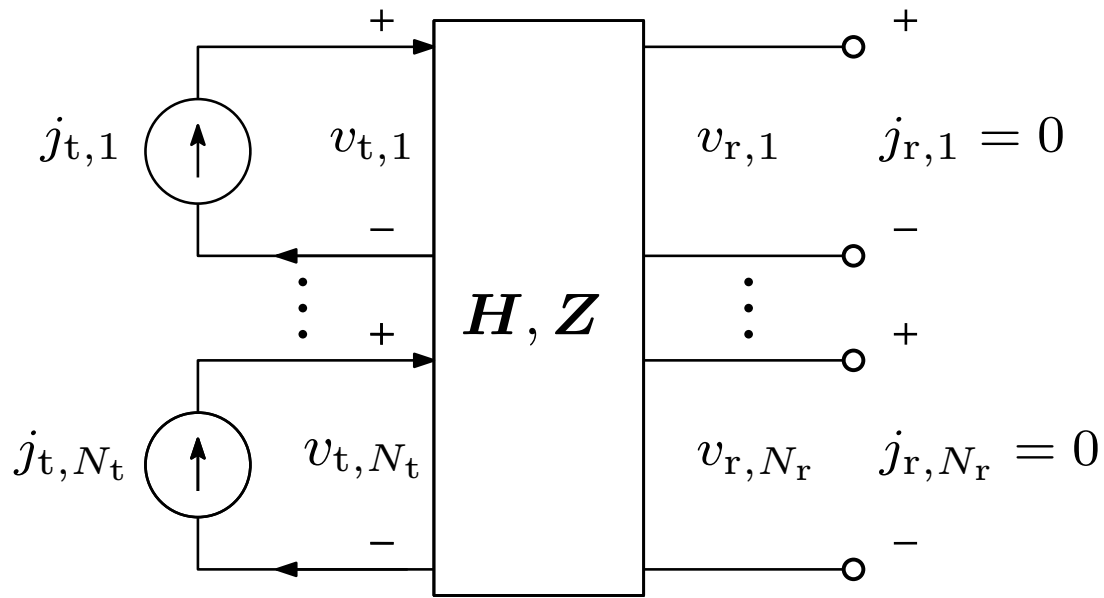
$$\mathbf{H}_Z = \mathbf{U} \tilde{\mathbf{H}}_Z \mathbf{V}^H$$

$$[\mathbf{U}]_{n,i} \approx \frac{e^{j \frac{2\pi}{N_r} i n}}{\sqrt{N_r}}, \quad [\tilde{\mathbf{H}}]_{i,\ell} \sim \mathcal{N}_{\mathcal{C}} \left(0, \frac{N_r N_t}{L_r L_t} \lambda_{i\ell}(Z) \right) \text{ IID}, \quad [\mathbf{V}^H]_{\ell,m} \approx \frac{e^{-j \frac{2\pi}{N_t} \ell m}}{\sqrt{N_t}}$$

$$\lambda_{i\ell}(Z) = L_r L_t \int_{f_i}^{f_{i+1}} \int_{\xi_\ell}^{\xi_{\ell+1}} S_Z(f, \xi) df d\xi$$

Exact UIU Model with Coupled Antennas

- Coupling appears as a **matrix multiplication** in the exact MIMO model^{1,2}
- Corresponds to ideal signal generators and infinite loads
- As **L/λ increases**, the UIU model asymptotically converges to the Fourier model



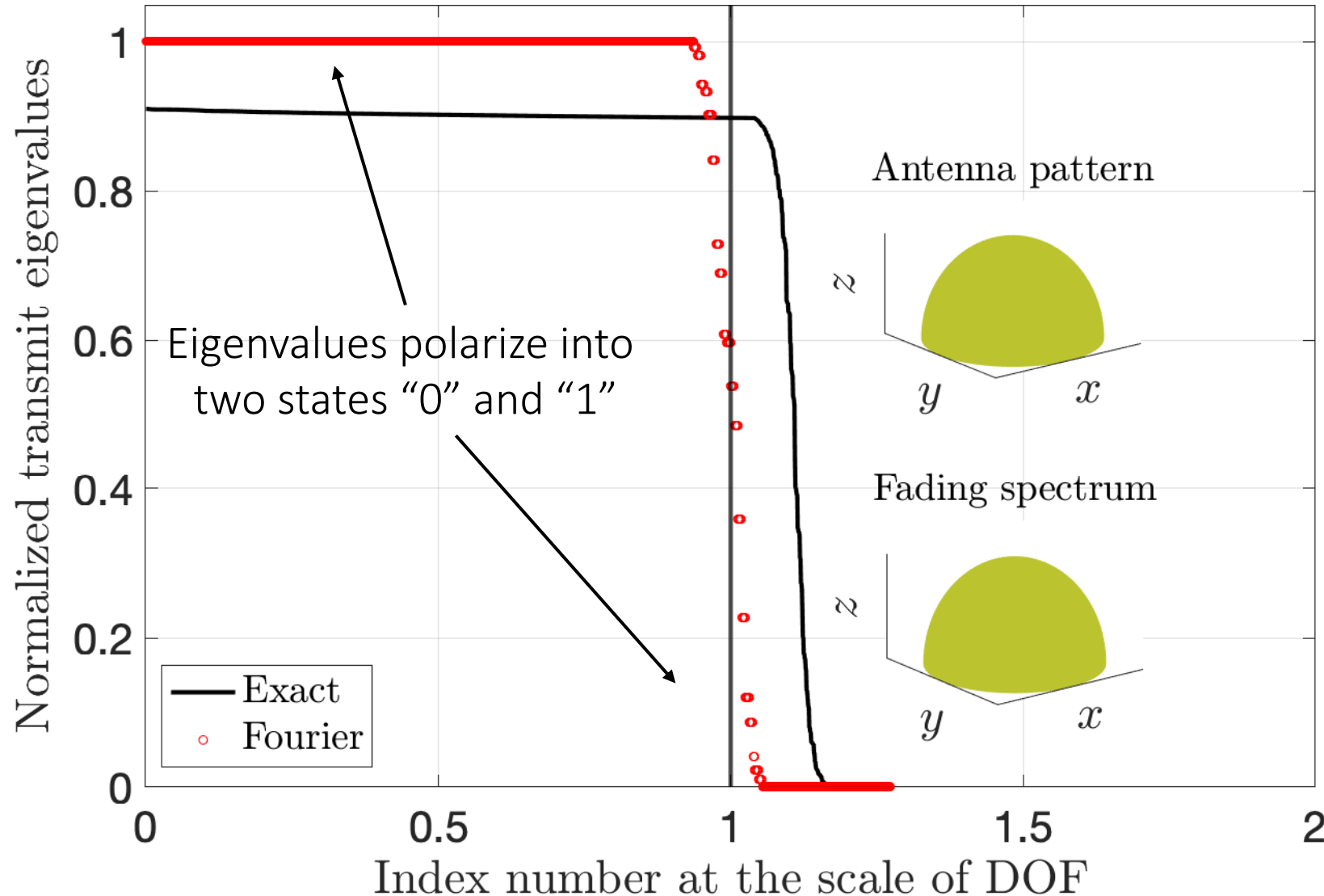
$$\mathbf{H}_Z = \mathbf{Z}^{-1/2} \mathbf{H}$$

$$[\mathbf{Z}]_{n,i} = \int_{-1/\lambda}^{1/\lambda} Z(f) e^{i2\pi f(x_{t,n} - x'_{t,n})} df$$

[1] J. Wallace and M. Jensen, "Mutual coupling in MIMO wireless systems: A rigorous network theory analysis," IEEE Trans. Wireless Commun., 2004.

[2] M. Ivrlac and J. Nosssek, "Toward a Circuit Theory of Communication," IEEE Trans. Circuit Systems, 2010

Fourier Model Accuracy with Finite Apertures



$$L_t/\lambda = 20$$

$$d_t/\lambda = 1/4$$

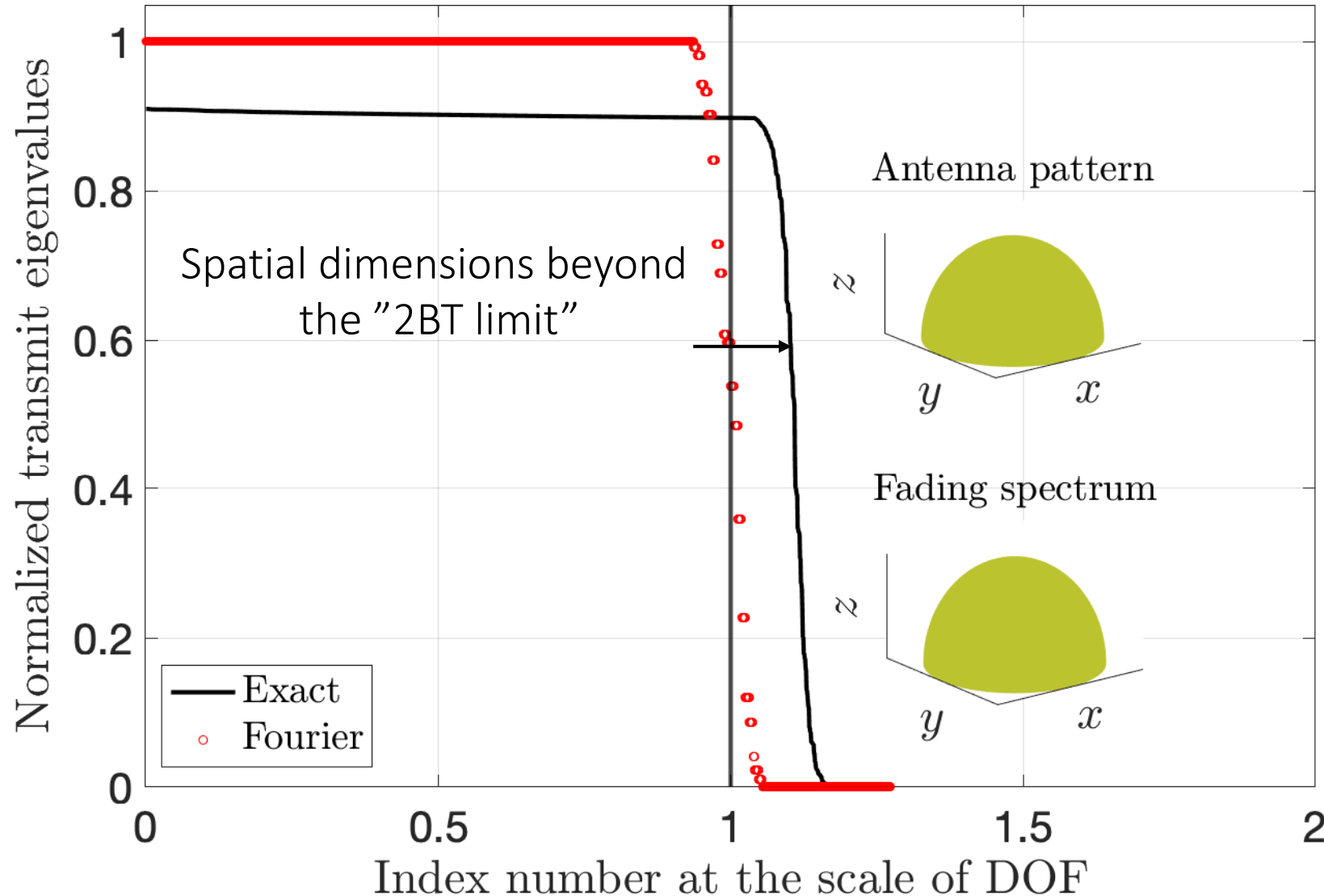
Separable scattering

Clarke – Jakes spectrum

Isotropic antenna pattern

$$\text{DOF} = \left\lceil \frac{2L_t}{\lambda} \right\rceil$$

Fourier Model Accuracy with Finite Apertures



$$L_t/\lambda = 20$$

$$d_t/\lambda = 1/4$$

Separable scattering

Clarke – Jakes spectrum

Isotropic antenna pattern

$$\text{DOF} = \left\lceil \frac{2L_t}{\lambda} \right\rceil$$

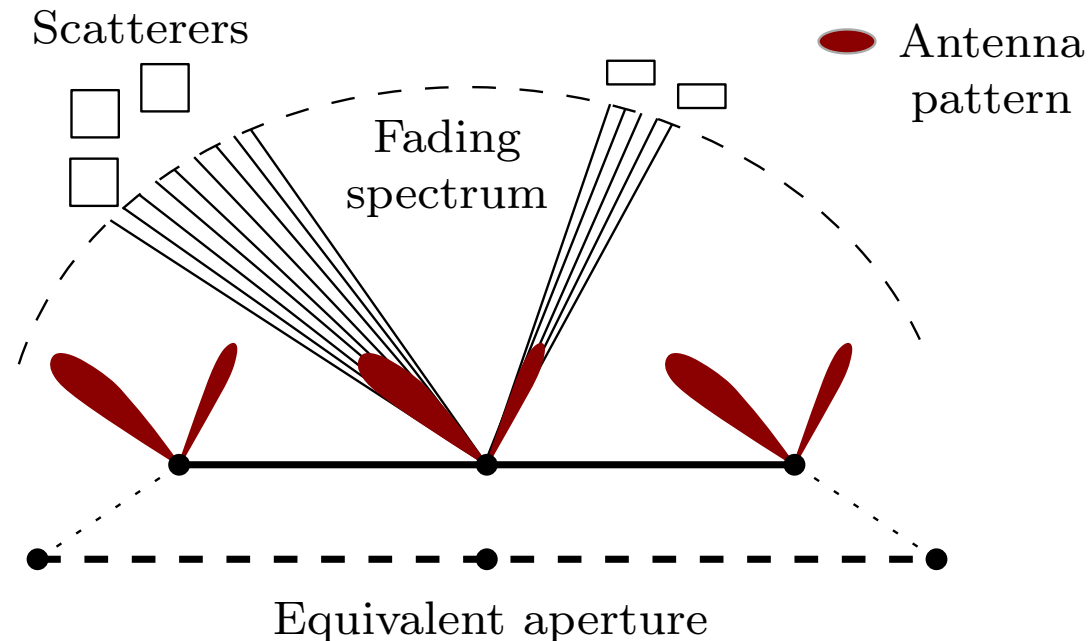
Reconfigurable Antenna Arrays

Adjusting Antenna Patterns to Fading Conditions

- Reconfigurable antenna patterns allow to tune the cascaded response through $A(f)$

$$S_Z(f) = \frac{E\{|H(f)|^2\}}{|A(f)|^2}, \quad |f| \leq 1/\lambda$$

- Matching antenna pattern to the fading flattens the spectrum, appears to reduce correlation
- Equivalent to spreading uncoupled antennas apart over an expanded aperture

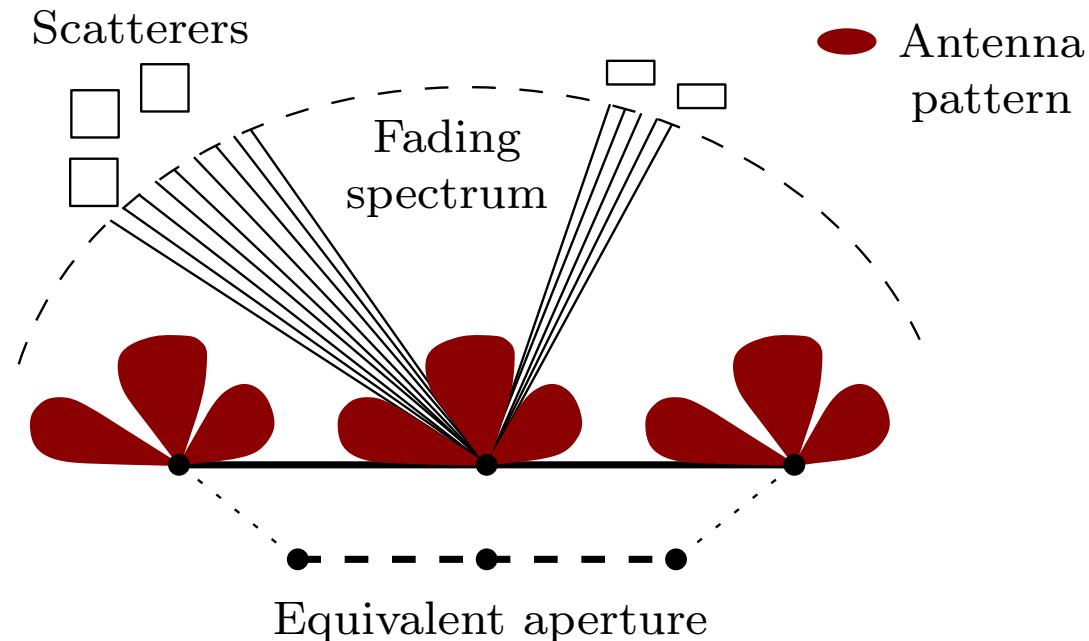


Adjusting Antenna Patterns to Fading Conditions

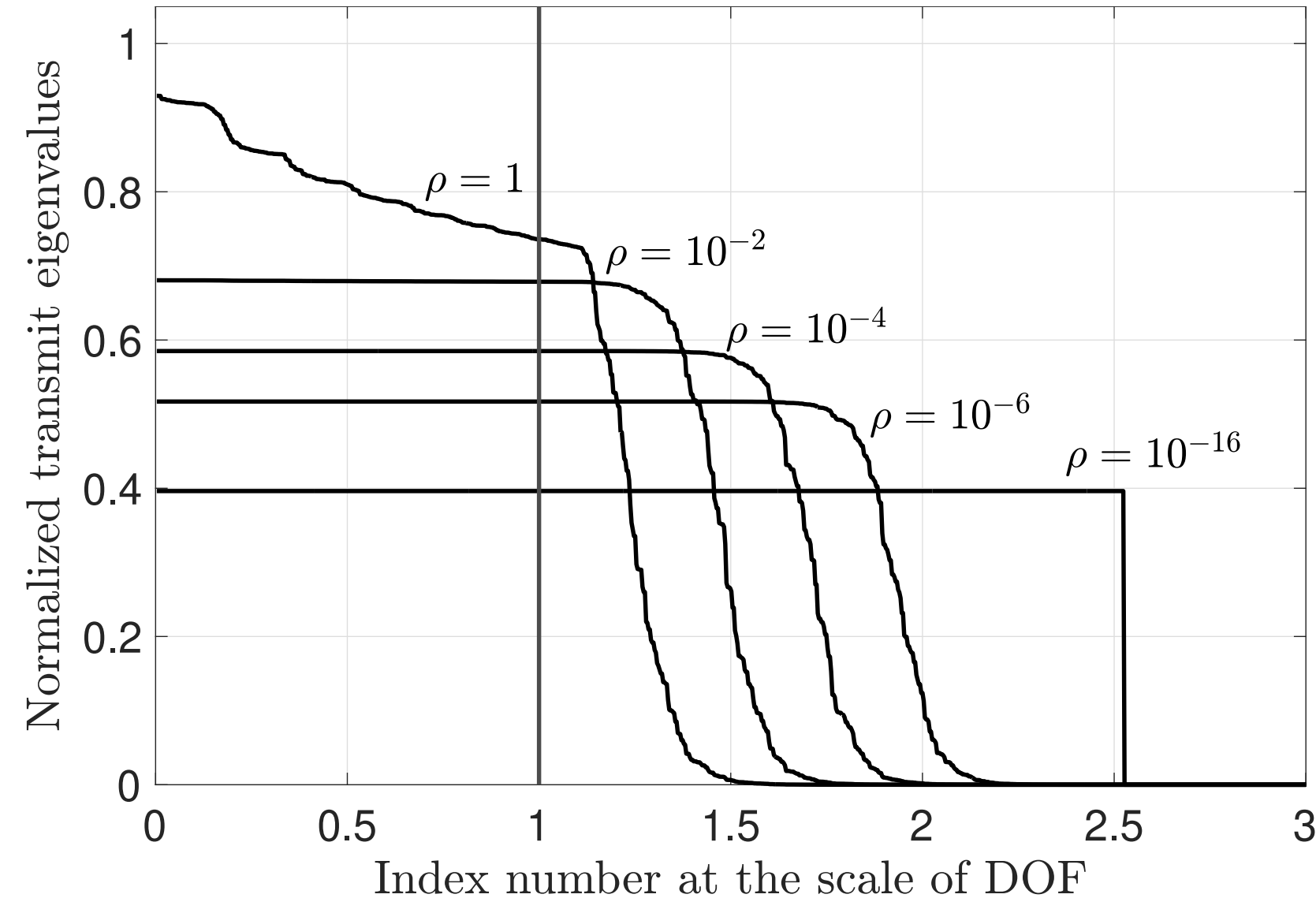
- Reconfigurable antenna patterns allow to tune the cascaded response through $A(f)$

$$S_Z(f) = \frac{E\{|H(f)|^2\}}{|A(f)|^2}, \quad |f| \leq 1/\lambda$$

- Countering antenna pattern to the fading peaks the spectrum, appears to enhance correlation
- Equivalent to drawing uncoupled antennas together over a contracted aperture



Beyond the “2BT Limit”: Impact of Ohmic Antenna Losses



$$L_t/\lambda = 10$$

$$d_t/\lambda = 1/4$$

Separable scattering

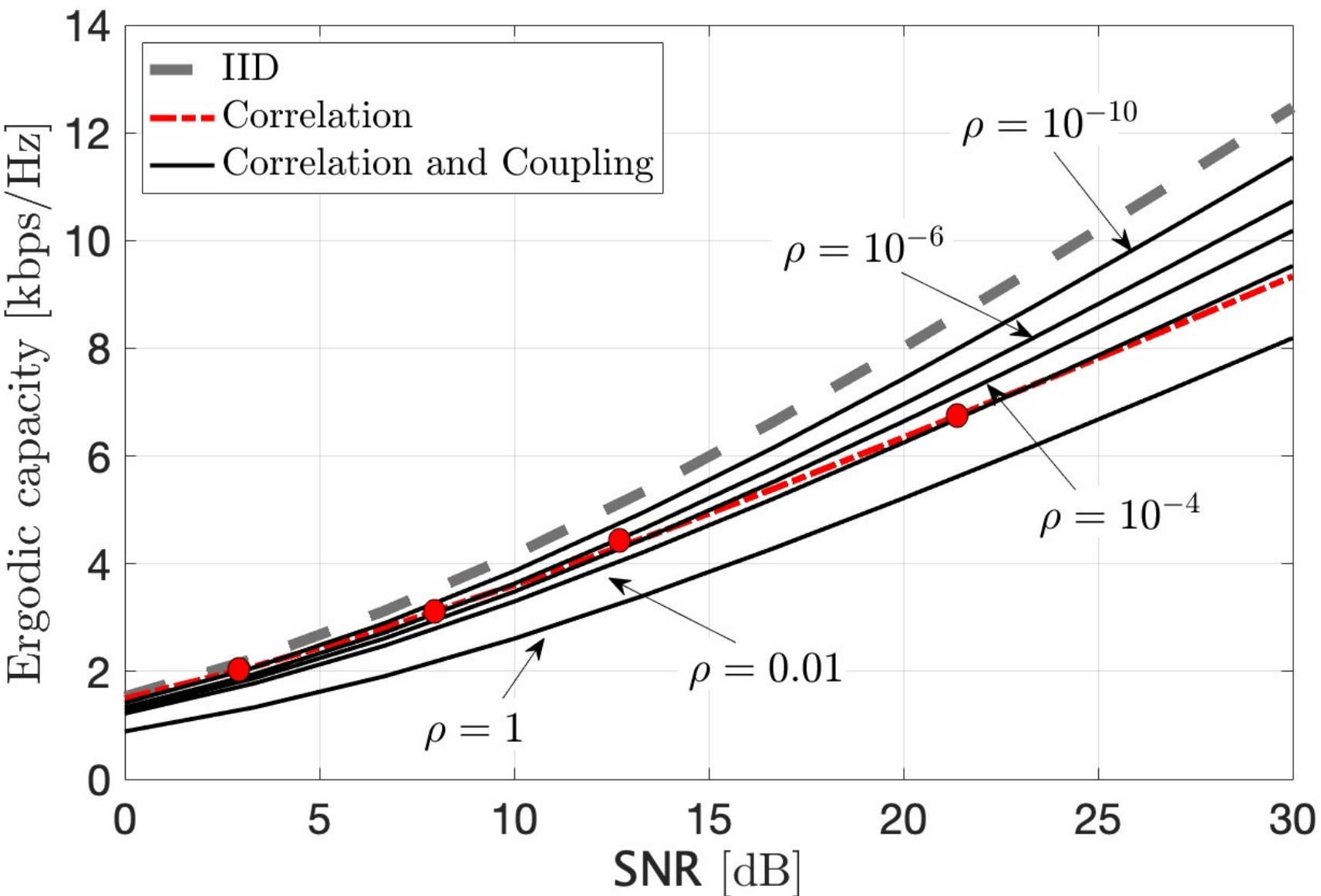
Clarke – Jakes spectrum

Isotropic antenna pattern

$$\text{DOF} = \left\lceil \frac{2L_t}{\lambda} \right\rceil$$

$$\mathbf{Z}(\rho) = \mathbf{Z} + \rho \mathbf{I}, \quad \rho : \text{ohmic losses}$$

Ergodic Capacity with Antenna Pattern Matched to Fading



$$L_t/\lambda = 15$$

$$d_t/\lambda = 0.4$$

Clarke – Jakes transmit spectrum

IID receive fading

Isotropic antenna pattern

$$\mathbf{Z}(\rho) = \mathbf{Z} + \rho \mathbf{I}, \quad \rho : \text{ohmic losses}$$

Conclusion

Take-Away Points

- **Signal theory** concepts (e.g., bandwidth, spectral representation) naturally extend to the electromagnetic domain
- Electromagnetic phenomena such as **multipath fading** and **mutual coupling** can be modeled as **linear filtering operations**
- Reconfigurable antenna patterns enable **manipulation of the cascaded filter response** to optimize performance under varying environmental conditions
- Dynamically adjusting the antenna patterns according to fading conditions **mimics an aperture expansion/contraction** (i.e., spatial dimensions increase/decrease)

Open Research Problems

- Design guidelines are derived for antenna pattern optimization in signal-to-noise-limiting regimes with further exploration needed for the **optimal strategy across signal-to-noise-ratios**
- Generalize fading and coupling models to **wideband scenarios**
- Analyze **pattern and polarization** diversity in non-stationary environments, where different parts of the array are exposed to distinct portions of the fading spectrum
- Evaluate the impact of **hardware impairments** introduced by impedance networks and realistic signal generators

Thank you for your attention!
Questions?